

■ WavyGo ERP System — Technical & Functional Documentation

Document Title: WavyGo Enterprise Resource Planning (ERP) System — Technical, Architecture & Functional Reference

Version: 2.0 (Updated & Codebase Aligned)

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Technology Stack: React.js (Frontend) + FastAPI (Backend) + MongoDB (Database)

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■ 1. Executive Summary & Document Overview

This document provides a comprehensive technical, functional, architecture, security, and schema reference for the **WavyGo ERP System**. It reflects the actual, operational codebase and replaces legacy or preliminary document specifications.

Core Objectives:

- **Centralize Operations:** Single source of truth for company management, task allocation, workforce attendance, performance reviews, EV rental marketplace operations, and business opportunities.
- **Granular Multi-Role Security:** Enforce role-based access control across **5 distinct roles**: **Founder**, **Admin**, **Manager**, **Employee**, and **Intern**.
- **Intelligent Collaboration:** Integrated real-time team messaging (**WavyGo Connect**) and intelligent AI assistance (**WavyGo AI**).
- **Auditable & Compliant:** Full activity stream logging, audit trail, and instant notifications.

■■ 2. Technology Stack & Architecture

2.1 Technology Stack Matrix

Layer	Technology / Library	Purpose & Implementation Details
Frontend Framework	React.js 18 + Vite	User interface, client-side routing (<code>react-router-dom</code>), component rendering.
Styling & UI Design	Tailwind CSS + Lucide Icons + Sonner	Responsive enterprise dashboard styling, toast alerts, custom design tokens.

Backend Framework	FastAPI (Python 3.10+)	High-performance async REST API, dependency injection for auth & RBAC.
Database Engine	MongoDB	Document-based NoSQL database for flexible enterprise schemas.
Database Driver	PyMongo / Motor (AsyncIOMotorClient)	Non-blocking asynchronous database driver for FastAPI.
Data Validation	Pydantic v2	Strict request/response payload schema validation & serialization (BaseDocument).
Authentication	JWT (JSON Web Tokens) + Refresh Sessions	Dual-token auth architecture with device IP, User-Agent & session revocation.
Password Hashing	passlib[bcrypt]	Cryptographic password hashing and verification.
Testing Suite	Pytest + Asyncio	Automated backend RBAC and endpoint unit testing (backend/tests/rbac_test.py).

2.2 System Architecture Diagram



■ 3. User Roles & Access Control (RBAC)

The WavyGo ERP platform supports **5 granular user roles**:

1. **■ Founder**: Organization owner with unrestricted access across all 17 modules, financial vaults, EV marketplace management, and company governance.
2. **■■■ Admin**: High-level operational administrator capable of managing employees, departments, task boards, announcements, and system settings.
3. **■ Manager**: Department or team leader empowered to create/assign tasks, review team leaves, issue performance reports, and monitor team activities.
4. **■ Employee**: Core team member focused on assigned tasks, personal workspace, attendance check-in/out, leave applications, and team chat.
5. **■ Intern**: Entry-level workspace user with focused task execution permissions and personal attendance logging.

3.1 Module Access Permission Matrix (MODULE_ACCESS)

Module Key	Sidebar Route	■ Founder	■■■ Admin	■ Manager	■ Employee	■ Intern
dashboard	/dashboard	■	■	■	■	■
marketplace	/marketplace	■	■	■	■	■
task-board	/task-board	■	■	■	■ (My Tasks)	■ (Assigned)
opportunity-hub	/opportunity-hub	■	■	■	■ (My Opps)	■
employees	/employees	■	■	■	■ (Workspace)	■ (Workspace)
wavygo-connect	/wavygo-connect	■	■	■	■	■
company-vault	/company-vault	■	■	■	■	■
finance	/finance	■	■	■	■	■
crm	/crm	■	■	■	■	■
marketing	/marketing	■	■	■	■	■
analytics	/analytics	■	■	■	■	■
calendar	/calendar	■	■	■	■	■
wavygo-ai	/wavygo-ai	■	■	■	■	■
about-wavygo	/about-wavygo	■	■	■	■	■
activity-logs	/activity-logs	■	■	■	■	■
notifications	/notifications	■	■	■	■	■
settings	/settings	■	■	■	■ (My Profile)	■ (My Profile)

3.2 Dynamic Contextual Sidebar Labels (SIDEBAR_LABELS)

To optimize user experience based on role context, the sidebar automatically renames modules:

- **task-board:**
 - Employee: "My Tasks"
 - Intern: "Assigned Tasks"
- **opportunity-hub:**
 - Employee: "My Opportunities"
- **employees:**
 - Employee & Intern: "My Workspace"
- **settings:**
 - Employee & Intern: "My Profile"

3.3 Granular Action Permissions (ACTIONS)

Action Domain	Permission Action Key	Permitted Roles
User Governance	<code>user.invite.admin</code>	Founder
	<code>user.invite.manager / employee / intern</code>	Founder, Admin
	<code>user.delete</code>	Founder
	<code>user.edit_others</code>	Founder, Admin, Manager
	<code>user.edit_self</code>	All Roles
Marketplace	<code>marketplace.any</code>	Founder Only
Tasks	<code>task.create</code>	Founder, Admin, Manager, Employee
	<code>task.assign / task.edit_any</code>	Founder, Admin, Manager
	<code>task.edit_own / task.comment</code>	All Roles
	<code>task.delete</code>	Founder, Admin
Opportunities	<code>opportunity.create / opportunity.assign</code>	Founder, Admin, Manager
	<code>opportunity.edit_any</code>	Founder, Admin
	<code>opportunity.edit_own</code>	Founder, Admin, Manager, Employee
	<code>opportunity.delete</code>	Founder
Workforce	<code>employee.invite / department.create</code>	Founder, Admin

	<code>attendance.mark_self / leave.request_self</code>	All Roles
	<code>attendance.mark_others / leave.approve</code>	Founder, Admin, Manager
	<code>performance.create</code>	Founder, Admin, Manager
WavyGo Connect	<code>connect.create_channel</code>	Founder, Admin, Manager
	<code>connect.create_announcement</code>	Founder, Admin
	<code>connect.send_dm</code>	All Roles
Audit & Config	<code>activity.view_all</code>	Founder, Admin
	<code>activity.view_team</code>	Manager
	<code>settings.company_edit</code>	Founder

■ 4. Core ERP Modules & Capabilities

■ 4.1 EV Rental Marketplace & Operations Hub (Founder Only)

- **City Operations:** Active, paused, or planned deployment across target cities.
- **Vendor Management:** Vendor directory, contact details, rating, and KYC verification status.
- **Fleet Management:** Vehicle tracking (bikes, scooters, ebikes), registration plates, status (`available`, `booked`, `maintenance`, `retired`), and hourly/daily rental tariffs.
- **Customer & Booking Engine:** Customer profiles, booking start/end timestamps, revenue calculations, pricing tiers, and promotional discount coupons.
- **Compliance & Support:** KYC document uploads (Aadhaar, PAN, DL, GST, CIN), support ticket triage, customer/vendor ratings and reviews.

■ 4.2 Task & Project Board

- **Task Lifecycle:** `todo` → `in_progress` → `review` → `completed` (or `cancelled`).
- **Priority Levels:** `low`, `medium`, `high`, `urgent`.
- **Granular Fields:** Assignee, reporter, module tag, due date, attachments, external links.
- **Subtasks & Comments:** Nested subtasks with checkbox completion state; interactive comment feeds with file attachments.

■ 4.3 Workforce Management (Employees, Attendance & Leaves)

- **Employee Directory:** Full list of organization staff, designations, departments, contact info, and online statuses.

- **Attendance Tracker:** One-click check-in/out logging, automatic shift duration calculation, and status tags ([present](#), [absent](#), [leave](#), [half_day](#), [wfh](#)).
- **Leave Approval Workflow:** Employee leave requests ([casual](#), [sick](#), [earned](#), [unpaid](#)), reason logs, and Manager/Admin approval/rejection interface.
- **Performance Evaluation:** Quarterly/annual evaluation ratings (1–5 scale), key accomplishments, and growth area reviews.

■ 4.4 Business Opportunity Hub

- **Opportunity Pipeline:** Track grants, investors, accelerators, incubators, tenders, CSR initiatives, partnerships, workshops, and conferences.
- **Financial Value & Assignee:** Track opportunity value in Lakhs (₹), submission deadlines, assigned owners, document attachments, and pipeline status ([open](#), [assigned](#), [in_progress](#), [won](#), [lost](#), [closed](#)).

■ 4.5 WavyGo Connect (Communication Hub)

- **Channel Types:** Public channels, Direct Messages (DMs), group chats, and organization-wide announcements.
- **Real-Time Messaging:** Threaded messages, attachment sharing, member rosters, and instant unread notification indicators.

■ 4.6 WavyGo AI Assistant

- Integrated enterprise AI copilot offering instant responses to operational queries, workflow guidance, task assistance, and metric summaries.

■ 4.7 Audit Logs & Notifications

- **Activity Logs:** Immutable record of actions containing [user_id](#), [user_name](#), [user_role](#), [action](#), [module](#), [target](#), metadata, and creation timestamp.
- **Notification System:** User-targeted or broadcast alerts ([info](#), [success](#), [warning](#), [error](#)) with direct navigation links.

■■ 5. MongoDB Data Model Schema

The database consists of the following primary MongoDB collections:

wavygo_db	
■■■ users	# User credentials, roles, profile info, online status
■■■ sessions	# JWT refresh token sessions & revocation tracking
■■■ departments	# Department structure & department head IDs
■■■ tasks	# Task details, assignees, subtasks, deadlines

■■■ task_comments	# Comments & file attachments linked to tasks
■■■ attendance	# Daily check-in/out records & status
■■■ leaves	# Employee leave submissions & review status
■■■ performance	# Performance reviews, scores, & feedback
■■■ opportunities	# Business opportunities, grants, value, assignees
■■■ connect_channels	# Channels, DMS, & member list schemas
■■■ connect_messages	# Real-time chat messages & attachments
■■■ marketplace_cities	# Active/planned EV rental operation cities
■■■ marketplace_vendors	# Vendor profiles & KYC verification state
■■■ marketplace_vehicles	# Vehicle fleet, pricing, & maintenance status
■■■ marketplace_customers	# Customer profiles & KYC status
■■■ marketplace_bookings	# Customer booking contracts & revenue
■■■ marketplace_pricings	# Hourly/daily/weekly/monthly pricing plans
■■■ marketplace_coupons	# Promotional discount coupons & usage limits
■■■ marketplace_kyc	# Vendor & customer compliance documentation
■■■ marketplace_support	# Operational support tickets & status
■■■ marketplace_reviews	# Customer & vendor review scores
■■■ notifications	# System notifications & broadcast messages
■■■ activity_logs	# Organization audit trail

■ 6. Authentication & Security Architecture

1. **Password Hashing:** Stored strictly using `bcrypt` hashes via `passlib`. Plaintext passwords are never logged or stored.
2. **Dual JWT Token Architecture:**
 - **Access Token:** Short-lived JWT bearer token passed in the `Authorization: Bearer` header for protected API routes.
 - **Refresh Token:** Stored in the `sessions` collection with user-agent, IP address, creation timestamp, and revocation boolean state.
3. **Backend Authorization Enforcement:**
 - API endpoints use FastAPI dependency injection (`get_current_user`, `require_roles("Founder", "Admin")`, `can(role, action)`).
 - Frontend UI hidden elements are strictly backed by server-side permission rejection (`HTTP 403 Forbidden`).
4. **Input Sanitization & NoSQL Protection:**
 - Pydantic v2 schemas reject malformed fields or direct query object injections.
5. **CORS & Environment Confidentiality:**
 - Strict CORS origins (`CORS_ORIGINS`). Database connection strings and JWT secrets are managed via `.env`.

■ 7. REST API Endpoints Reference

Module	Method	Endpoint Path	Description & Access
Auth	POST	<code>/api/auth/login</code>	Authenticate user & issue tokens

	POST	/api/auth/register	Register new user (Founder/Admin only)
	POST	/api/auth/refresh	Issue new access token using refresh token
	POST	/api/auth/logout	Revoke session & clear online status
	GET	/api/auth/me	Fetch authenticated user profile
Users	GET	/api/users/directory	Fetch directory of active users
Dashboard	GET	/api/dashboard/stats	Fetch executive KPIs & role analytics
Employees	GET	/api/employees/directory	View employee list & departments
	POST	/api/employees/invite	Invite employee (Founder/Admin)
	POST	/api/employees/attendance/check-in	Log daily check-in timestamp
	POST	/api/employees/attendance/check-out	Log daily check-out timestamp
	GET / POST	/api/employees/leaves	List or submit leave requests
	PUT	/api/employees/leaves/{id}	Approve or reject leave request
Tasks	GET / POST	/api/tasks	List or create tasks
	PATCH	/api/tasks/{id}/status	Update task workflow status
	POST	/api/tasks/{id}/comments	Add comment thread to task
Opportunities	GET / POST	/api/opportunities	List or register business opportunities
	PUT	/api/opportunities/{id}/assign	Assign opportunity to employee
Connect	GET / POST	/api/connect/channels	List or create chat channels
	GET / POST	/api/connect/messages	List or post channel messages / DMs
Marketplace	GET / POST	/api/marketplace/*	Manage cities, vendors, vehicles, bookings (Founder)
Activity	GET	/api/activity	Retrieve system audit logs
Notifications	GET / PATCH	/api/notifications	Fetch alerts or mark as read

■ 8. Testing & Quality Assurance

To verify system functionality and RBAC policy enforcement, run the automated test suite:

```
cd backend
pytest backend/tests/rbac_test.py
```

Verified Test Scenarios:

- **■ Authentication:** Valid login, invalid credentials rejection, password hashing check.
 - **■ RBAC Access Control:** Founder-only endpoint access vs Employee 403 Forbidden checks.
 - **■ Task Lifecycle:** Creation, assignment, status state changes, and comment posting.
 - **■ Attendance & Leaves:** Check-in/out timestamp validation and leave review approvals.
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■ 9. Future Scope & Roadmap

1. **WebSockets for WavyGo Connect:** Upgrade real-time chat from polling to full duplex WebSockets.
 2. **Automated AI Task Scheduling:** AI auto-assignment based on workload & department availability.
 3. **Advanced EV Fleet Telematics:** Real-time GPS and IoT telemetry integration for vehicles.
 4. **Mobile Native Applications:** iOS and Android mobile wrappers for field attendance and notifications.
 5. **SSO & Two-Factor Authentication (2FA):** Enterprise OAuth2 / SAML integration.
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WavyGo ERP System — Empowering workforce efficiency, intelligent automation, and company growth.